MMS

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October 2026

1 Introduction

To implement the method of manufactured solutions (MMS) in 1-D, we choose the following separable solution:

$$\psi_g(x,t) = \frac{c}{4\pi} B_g \left(1 + e^{-\tau t/2}\right) f(x), \tag{1}$$

$$T_{\text{rad}}(x,t) = T_0 \left[\left(1 + e^{-\tau t/2}\right) f(x)\right]^{1/4}, \tag{2}$$

$$T_{\text{mat}}(x,t) = T_0 \left[\left(1 - e^{-\tau t/2}\right) f(x)\right]^{1/4}. \tag{3}$$

This is a separable form of ψ_g that allows us to utilize any desired spatial distribution. In this case, we choose a Gaussian:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}. \tag{4}$$

The governing equations of the MMS system are

$$\frac{1}{c}\frac{\partial \psi_{g,m}}{\partial t} + \mu_m \frac{\partial \psi_{g,m}}{\partial x} = \sigma_{a,g}(T)\left(4\pi B_g(T) - \psi_{g,m}\right) + S_{g,m}, \tag{5}$$

$$C_v \frac{\partial T}{\partial t} = \sum_{g=1}^{G} \sigma_{a,g}(T)\left(4\pi\psi_g(\mu) - 4\pi B_g(T)\right) + Q. \tag{6}$$

To begin solving we take the necessary derivatives

$$\partial_t(\psi_g) = \frac{c}{4\pi} f(x) \left[\partial_t(B_g)\left(1 + e^{-\tau t/2}\right) - \frac{\tau}{2} B_g e^{-\tau t/2}\right], \tag{7}$$

$$\partial_x(\psi_g) = \frac{c}{4\pi}\left(1 + e^{-\tau t/2}\right)\left(B_g f'(x) + f(x) B_g'(x)\right), \tag{8}$$

$$f'(x) = -\frac{x}{\sigma^2} f(x), \tag{9}$$

$$\partial_t(T) = T_0 f(x)^{1/4} \frac{\tau}{2} e^{-\tau t/2}, \tag{10}$$

$$\partial_x(T) = \frac{1}{4} T^{-3}\left(1 - e^{-\tau t/2}\right) f'(x). \tag{11}$$

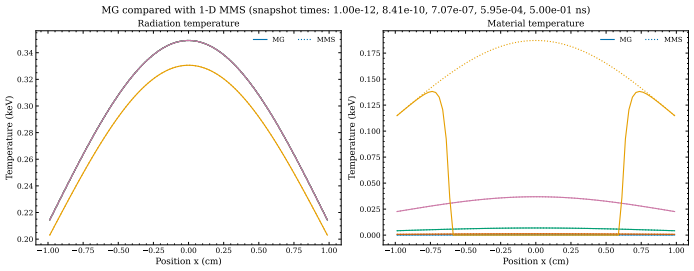


Figure 1: 200 Group Multigroup MMS

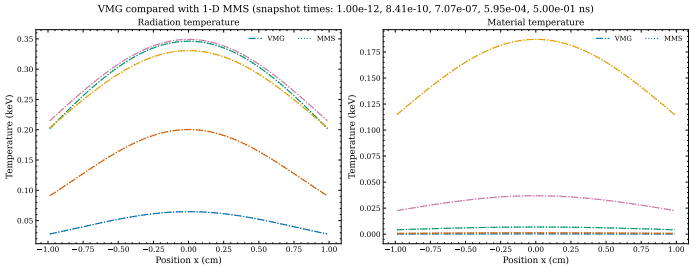


Figure 2: 200 Group VMG MMS

This gives all necessary equations to find S and Q omitting the derivatives of B_g due to the differences in definition between methods.